S3 Supplementary QC and Robustness

H5 coverage, modality-source checks, and regional MNPS audit

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Short title: S3 QC - MEEG concordance

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S3.0 Sample and epoch coverage overview

[The audit found 108 participant-run H5 files from 18 Healthy participants. All global 3D and 9D arrays were finite; median run length was 244 time points. The complete per-run audit is `results/eeg_meg_ds003645_20260723/00_sanity_check/run_coverage_and_regional_audit.csv`.]

S3.1 Modality source coverage

[All exported windows have both modality-specific feature surfaces available in the paired H5 contract. `row_source` flags make clear that individual rows originate from modality-native source files; paired feature matching is performed on aligned exported time grids, not by declaring every original row a simultaneous sensor sample.]

S3.2 Regional MNPS collapse

[For every audited run, all six EEG `/regional_mnps/<topology>/mnps` arrays were equal to one another within numerical tolerance. In contrast, no MEG run showed a collapsed six-region set. This is an export limitation rather than biological evidence of EEG regional homogeneity. The article excludes the collapsed EEG regional MNPS arrays from regional geometric and reachability tests.]

S3.3 Robustness summary

[The same 18 participants contribute to each matched feature test. Temporal correlations use finite concurrent samples; participant means aggregate run-level values. Wilcoxon and sign-flip permutation tests provide two independent paired tests, and BH correction is applied by global versus regional family.]

S3.5 Reviewer-facing robustness summary

Robustness summary

[The report supports feature-level EEG–MEG concordance and feature-mean differences. It does not support a claim of EEG–MEG concordance for regional MNPS geometry until the collapsed EEG export surface is repaired.]

Bibliography